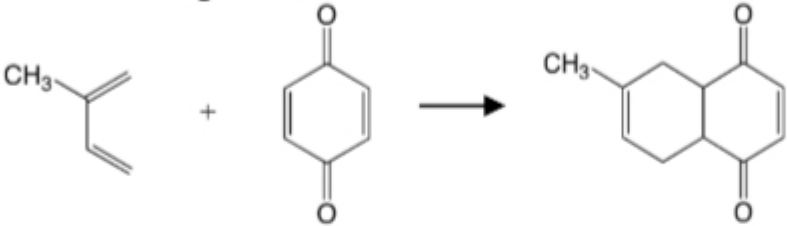
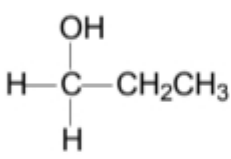
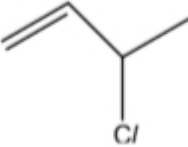
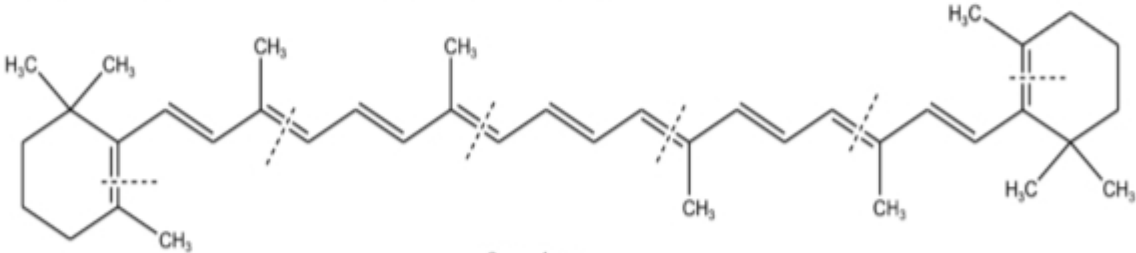


11 Alkenes

Answers to Multiple Choice Questions

1	2	3	4	5	6	7	8	9	10
A	B	A	C	D	B				

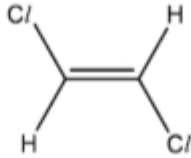
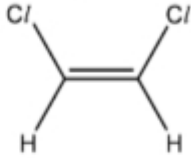
1	A Pattern recognition: 
2	B  is the minor product from the addition reaction due to the lower stability of the carbocation intermediate.
3	A As seen from the example, hydroboration is an anti-Markovnikov electrophilic addition reaction. It means a hydrogen atom is added to the more substituted carbon atom instead of the less substituted carbon atom. As such, subjecting but-1-ene to hydroboration should result in the formation of butan-1-ol instead of butan-2-ol.

4	C A is correct as slow step has one mole each of two reactants. B is correct as electron delocalization occurs with the carbocations changing position. ✓ C is not correct. The electrophile $\delta^+H-Cl^{\delta-}$ will add such that the more electron deficient $H^{\delta+}$ will be attached to the terminal position to get the more stable secondary carbocation.  The product should be D is correct; iodine atom carries the partial positive charge $\delta^+I-Cl^{\delta-}$ and is added before Cl.
5	D Moles furfuryl thiol is $0.10/114 = 8.77 \times 10^{-4}$ D moles of SO_2 produced is $21/24000 = 8.75 \times 10^{-4}$; moles of product with 4 Br added to double bonds $(0.38/433.6) = 8.76 \times 10^{-4}$ 1 is wrong as CO_2 produced is $0.23/44 = 0.00523$ mol; it should be $5 \times 8.77 \times 10^{-4}$ 3 is wrong as H_2O produced is $0.16/18 = 0.00889$ mol; it should be $3 \times 8.77 \times 10^{-4}$
6	B At the 6 cleavage points below (5 other points are not indicated), ketone will be produced; two of the products (from cleavage at the first and last point) contain two ketones in the same molecule. In total 6 molecules have ketones.  β-carotene

Answers to Structured and Free Response Questions

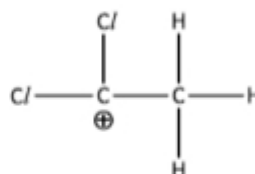
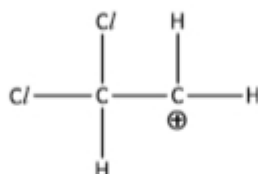
7

(ii) (i)

isomer	A	B
structure		
cis or trans isomer	trans	cis

(ii) The presence of the π -bond restricts the rotation of the molecule and two different groups attached to each carbon of the C=C bond.

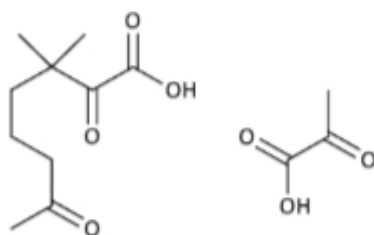
(b) (i)



(ii) The intermediate for I is more stable compared to II as the positively charged carbon is less destabilised by the 2 electron withdrawing chlorine substituents attached to the positively charged carbon. Hence I is the preferred product.

8

(i)

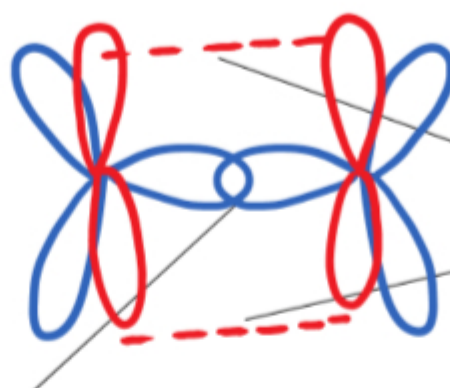


(ii) BCO is an enzyme and has a very specific action, in this case, to cleave only one of the C=C bonds. There are many other C=C double bonds in β -carotene, which will all be cleaved using acidified KMnO_4 in the laboratory.

9

PrelimNYJC17/III/2a, b

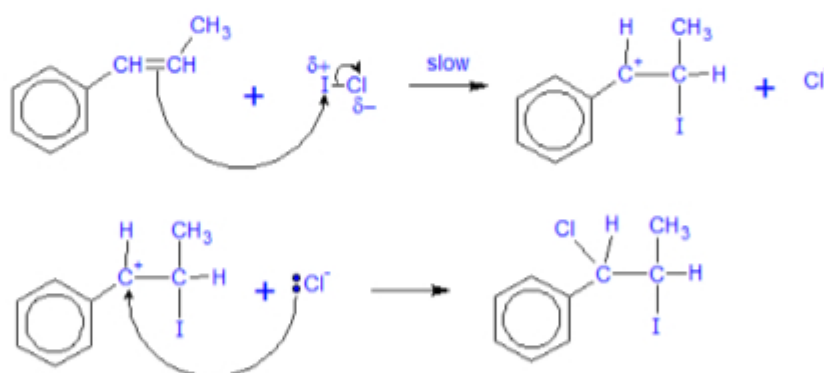
(a)



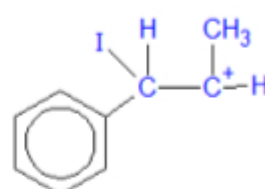
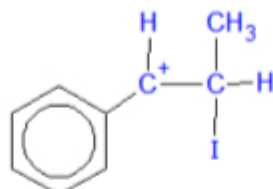
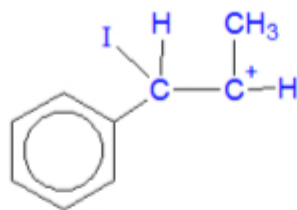
The unhybridised p-orbital overlap sideways to form π bond.

The sp^2 hybrid orbital overlap head on to form σ bond.

(b) (i) Electrophilic Addition

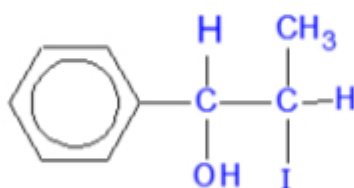


(ii)



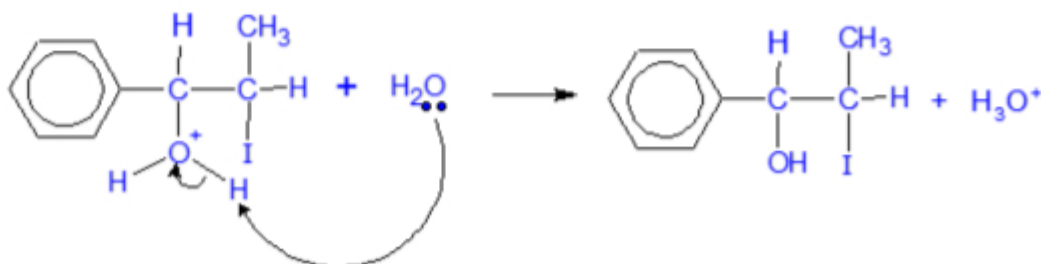
is more stable than and hence is the preferred carbocation formed. The empty p orbital of the carbocation centre overlaps with the delocalised π electron cloud of the benzene ring, electrons from the benzene ring delocalises to the carbocation centre. Hence the positive charge can be dispersed into the benzene ring, stabilizing it further.

(iii)



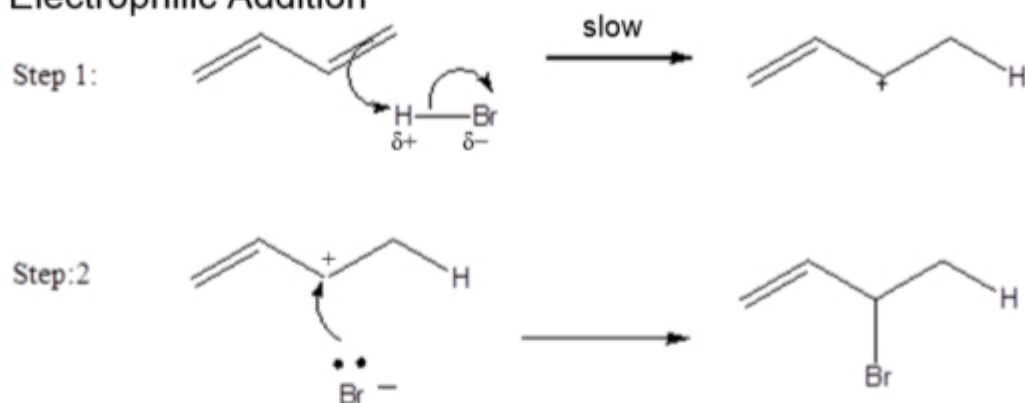
The lone pairs of electrons on O in H_2O acts as competing **nucleophile** in the 2nd step of the mechanism to attack the carbocation intermediate.

Note: A proton will be lost subsequently to form the organic compound as shown in the structure above.

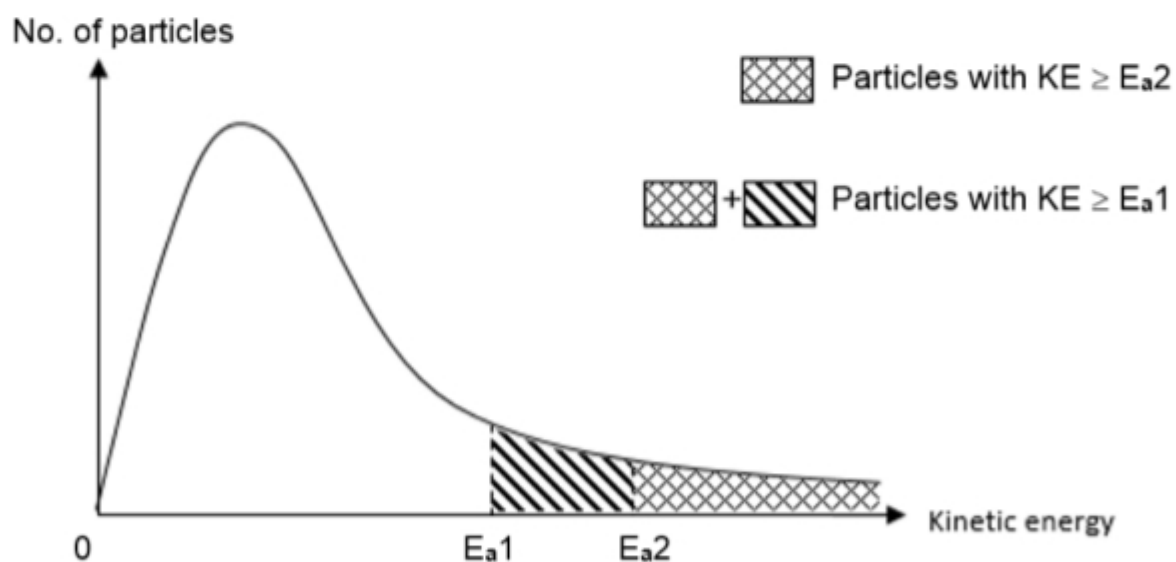


10 PrelimNJC17/III/4a, b

(a) (i) Electrophilic Addition



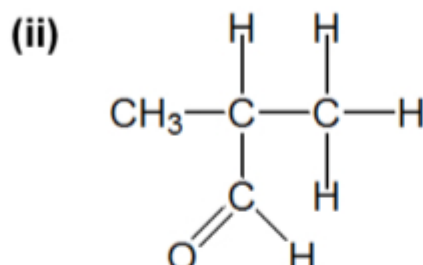
(b) (i)



	E_{a1} is the E_a for the formation of 3-bromobut-1-ene E_{a2} is the E_a for the formation of 4-bromobut-2-ene Axis labelled Peak of graph should be before plateau off Graph plateau off Graph starts from origin Legend for shaded area (ii) The activation energy for the formation of the 3-bromobut-1-ene is lower, hence there are more particles with energy larger than the activation energy. This leads to an increase in the frequency of effective collision and hence the rate of forming 3-bromobut-1-ene is faster. (iii) 4-bromobut-2-ene is a more substituted alkene hence it is more stable.							
11	<u>PrelimNJC17/III/4c</u> KMnO₄ (aq), H₂SO₄ (aq), heat <table border="1"> <tr> <td>L</td><td> Purple KMnO₄ decolourises Effervescence observed. The gas forms white ppt in limewater </td><td> </td></tr> <tr> <td>M</td><td>Purple KMnO₄ decolourises</td><td> </td></tr> </table>		L	Purple KMnO ₄ decolourises Effervescence observed. The gas forms white ppt in limewater		M	Purple KMnO ₄ decolourises	
L	Purple KMnO ₄ decolourises Effervescence observed. The gas forms white ppt in limewater							
M	Purple KMnO ₄ decolourises							

12 PrelimRVHS17/III/5ai, ii

- (i) C1: +1
C2: -2
C4: -3

13 PrelimDHS17/III/Q2(b)

(i)

	bonds broken	bonds formed	$\Delta H / \text{kJ mol}^{-1}$
step 1	1 C=C	1 C-C 1 C-I	$(-350 - 240) + 610 = +20$
step 2	1 H-Cl	1 C-H	$-410 + 431 = +21$

- (ii)
- Only the free radical addition of HBr to an alkene is likely to occur since $\Delta H_{\text{step 1}}$, $\Delta H_{\text{step 2}}$ are all exothermic.
 - This reaction is however unlikely to occur if HCl and HI were used.
 - In each case, there is one propagation step which is endothermic — step 2 for HCl and step 1 for HI.
(Note that ΔS for step 1 is negative and for step 2 is ≈ 0 . Hence with a positive ΔH , ΔG will be positive and the forward reaction is thermodynamically unfavourable.)

(iii) $\text{CH}_3\text{CHICH}_3$

